



CP5046 Group Presentation

Kindergarten Learning Game Design

Counting · Ordering · Comparison · Matching · Sorting

Assessment 3 · Professor Insu Song

Bohong Wang, Weichao Sun, Chen Xianzhe, Haoyang Liu

Team Charter

Team roles and the game themes each member contributed



Bohong Wang

Game Designer / Scrum Coordinator

Math + ordering games

- Fruit counting
- Animal ordering
- Vehicle ordering



Weichao Sun

Team Leader / Game Designer

English comparison games

- Animal size
- Item size
- Food size



Chen Xianzhe

Designer / Tester

Music matching games

- Musicians
- Instruments
- Jazz matching



Haoyang Liu

Game Designer / Scrum Coordinator

Art sorting games

- Colour sorting
- Start with a letter
- Categorise by colour

Project Goals

The goal was to design a small portfolio of child-friendly KOPO learning games

How we developed the goals

- Used user stories and team reviews to clarify learning needs
- Kept instructions short for kindergarten learners aged 4-6
- Selected familiar content: fruits, animals, vehicles, food, items, music and colours
- Used Figma / FigJam for planning and KOPO for prototype screens

Specific design outcomes

- 12 prototype levels across four learning directions
- Large images, simple drag/drop or click interaction, and clear feedback
- Reusable screen structure to reduce confusion across levels

Team (4 members)



12 KOPO learning game levels

🍎 Counting
& ordering

🔍 Size
comparison

🎵 Music
matching

🚗 Colour
sorting



Figma / FigJam workspace evidence

Design Rationale

User needs, constraints, decisions and trade-offs behind the final designs

User needs & constraints

- Young learners need clear wording, large images, and simple interaction
- KOPO limits pushed us to avoid complicated game mechanics
- Short sprint time required reusable layouts and fast revision cycles

Design decisions & trade-offs

- Prioritised clarity over visual complexity
- Used familiar objects to reduce cognitive load
- Balanced fun, feedback, difficulty and feasibility

Need-driven design

Simple instructions
Visual examples
Clear feedback
Large touch areas

helps children understand quickly

Main trade-off

**More features
vs
clearer learning
flow**

Decision: keep mechanics simple and improve engagement through themes.



User stories and review notes shaped design decisions

Design Evolution through Iterations: Sprint Pilot 1A

Discovery work and user-story evidence

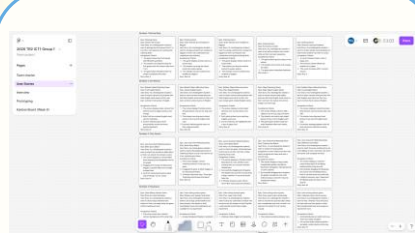
What we clarified first

- Collected user stories and learner assumptions
- Compared possible game types for KOPO feasibility
- Focused on simple activities for kindergarten learners

Pilot 1 converted broad ideas into a smaller, teachable design scope.



Team discussion



User story cards

Evidence used: team discussion photos, user story cards, assumptions and FigJam notes.

Main trade-off: many possible game ideas vs focused learning outcomes

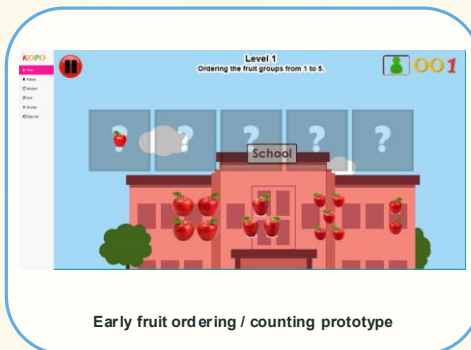
Design Evolution through Iterations: Sprint Pilot 1B

Early prototype scope and first interaction tests

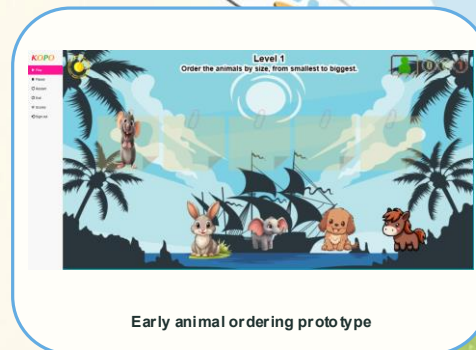
Prototype direction

- Built first ordering/counting screens as KOPO tests
- Checked whether images were large enough to understand
- Kept mechanics simple before adding more variety

Outcome: a reusable layout pattern for later levels.



Early fruit ordering / counting prototype



Early animal ordering prototype

Improvement direction: clearer instructions, fewer distractions, and more consistent visual structure.

Pilot 1 established the first working evidence before wider team production.

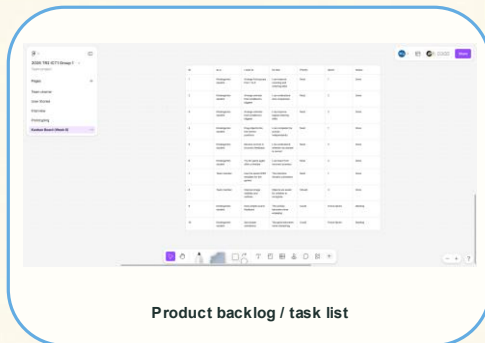
Design Evolution through Iterations: Sprint Pilot 2A

Backlog organisation and sprint planning

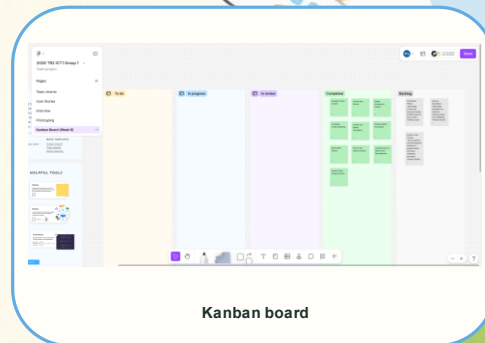
Planning activities

- Organised prototype tasks in the product backlog
- Used the Kanban board to track task status
- Assigned work across four learning themes

Planning helped the team move from ideas to deliverable prototype tasks.



Product backlog / task list



Kanban board

Main trade-off: more detailed planning vs faster implementation in a short sprint.

Pilot 2 made responsibilities and prototype progress easier to see.

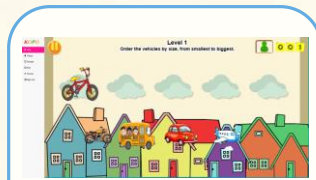
Design Evolution through Iterations: Sprint Pilot 2B

Prototype review and refinement decisions

Review focus

- Compared early screenshots across different levels
- Looked for layout and clarity issues
- Checked whether feedback and touch areas were obvious

The review connected prototype evidence to concrete design improvements.



Prototype review 1



Prototype review 2



Prototype review 3

Main improvement direction: clearer instructions, consistent layout, larger images and more obvious feedback.

Pilot 2 shifted the team from task completion to usability refinement.

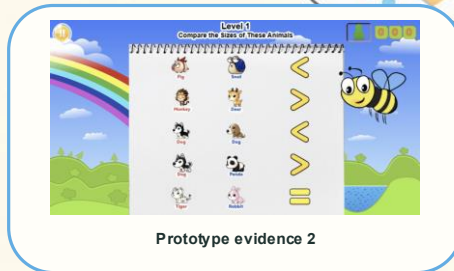
Design Evolution through Iterations: Sprint Pilot 3A

Group prototype evidence after feedback

Pilot 3 focus

- Applied review feedback to instructions and layouts
- Checked evidence from all members
- Compared image clarity and feedback presentation

Quality check before final consolidation.



Result: the team reviewed consistency across themes before retrospectives.

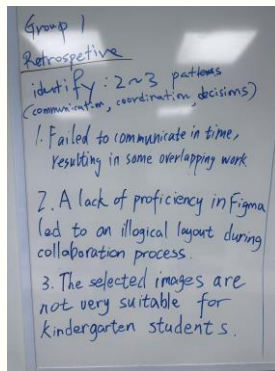
Design Evolution through Iterations: Sprint Pilot 3B

Sprint retrospective notes and follow-up actions

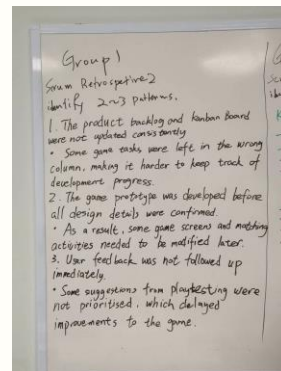
Retrospective findings

- Communication gaps affected update speed
- Backlog and Kanban needed stronger maintenance
- Some images and layouts needed clearer child-friendly presentation

Trade-off: final quality vs limited sprint time.



Retrospective 1: communication, layout, image suitability



Retrospective 2: backlog, Kanban, feedback follow-up

Follow-up: clarify task ownership earlier, update Kanban more consistently, and run stronger check-ins before final revisions.

Conclusion

Main outcomes, teamwork reflection, and future improvements

- Delivered a portfolio of 12 child-friendly KOPO learning game prototype levels
- Used sprint pilots to check designs against user stories, feedback and acceptance criteria
- Improved question clarity, image visibility, touch areas, feedback and layout consistency
- Teamwork was collaborative; roles, FigJam, Kanban and reviews supported coordination
- Possible improvements:
 - Clarify task ownership earlier in sprint planning
 - Hold stronger retrospectives and more frequent check-ins to identify blockers earlier

Final message: simple, familiar and consistent design is the strongest approach for young learners.